

A robotic arm with a gripper is reaching towards a human hand. The background is dark with colorful bokeh lights and abstract geometric shapes. The robotic arm is on the left, and the human hand is on the right. The gripper has the word 'ROBOTIQ' on it.

HUN
REN



23.09.2026 REVIVE WORKSHOP

Noemi Friedman:

SHM with Bayesian inversion and probabilistic models - Tools, Limitations and Challenges

HUN-REN
Hungarian Research Network

HUN-REN Institute for Computer Science and Control (HUN-REN SZTAKI)

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HUNGARIAN RESEARCH NETWORK INSTITUTE FOR COMPUTER SCIENCE AND CONTROL

(HUN-REN SZTAKI)



Established in 1964, the HUN-REN Institute for Computer Science and Control (SZTAKI) has nearly 300 full-time employees, 80 with Ph.D.

Since 2010, SZTAKI hosts the Fraunhofer-SZTAKI Project Centre for Production Management and Informatics (PMI). In 2018, the Fraunhofer Gesellschaft and HUN-REN SZTAKI established a common firm, the EPIC InnoLabs Nonprofit Ltd., to transfer the R&D&I results to the industry. The Institute is also the coordinator of the National Laboratory for Autonomous Systems.

The two National Laboratories collaborate closely and share the Innovation and Demonstration Space in the 11 Lágymányosi Street building.

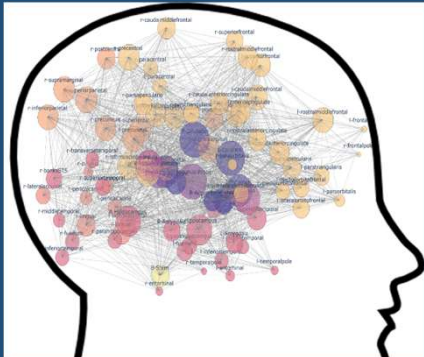
Nation-wide collaboration projects: Between 2020-2025, HUN-REN SZTAKI coordinates the **Artificial Intelligence National Laboratory** and the **Autonomous Systems National Laboratory**.

Compute facilities: Hungarian Research Cloud of 40 servers with 8 V100 or A100 GPUs each and available for our research.

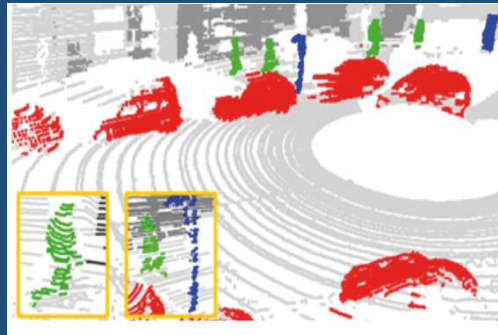
Exhibition space: HUN-REN SZTAKI hosts the over 400m2 Exhibition Space of the Artificial Intelligence National Laboratory Hungary, which is dedicated to demonstrations and promotion of AI technologies towards SMEs, large industry, as well as government agencies.



Six AI topics



Foundations



Machine perception



Human language tech



Medical, health



Sensor, IoT, Telecommunications



Security, Privacy, Infrsstructure

Motivation

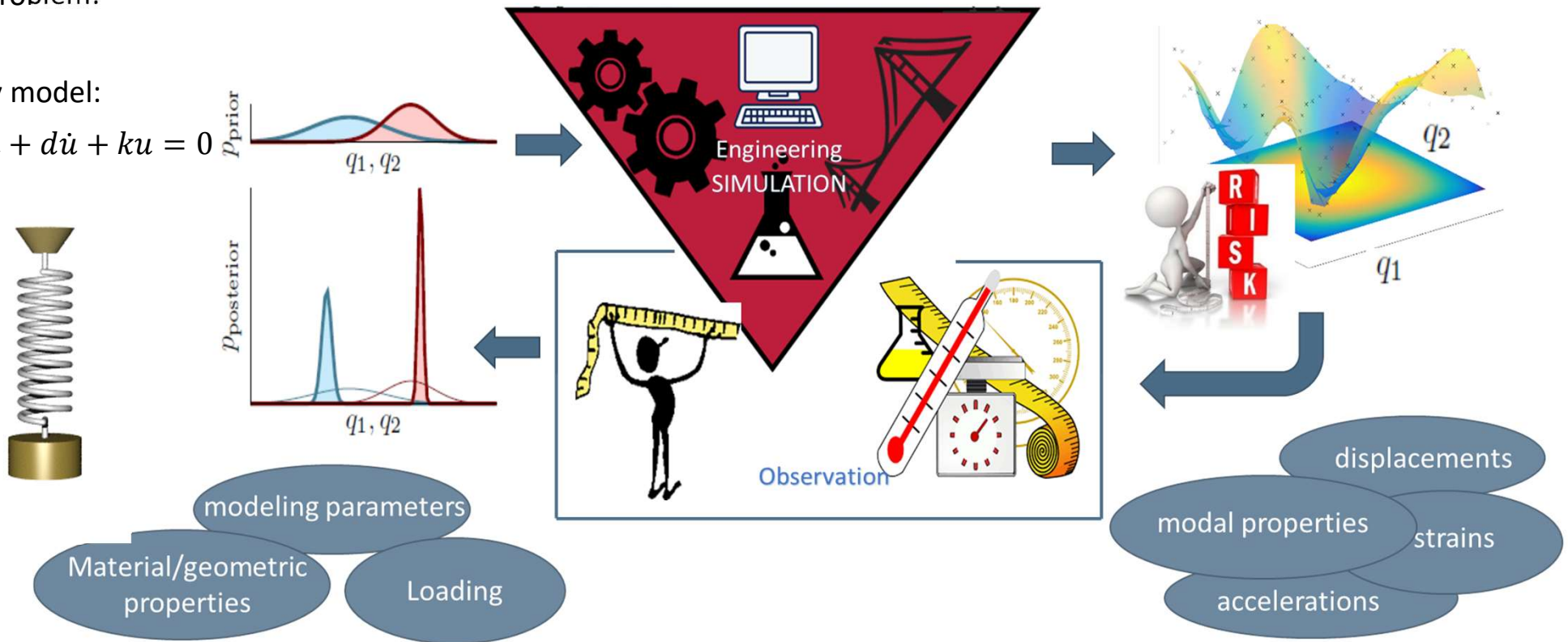


Motivation: Identifying material properties from SHM data

Inferring material/structural properties of an infrastructure from measured SHM data is typically described by an inverse problem.

Toy model:

$$m\ddot{u} + d\dot{u} + ku = 0$$



Motivation: Identifying material properties from SHM data



Implicit abstract description of the ODE/PDE:

$$\mathcal{A}(u(\omega), q(\omega)) = f(q(\omega))$$

The measurable QoI:

$$y = H(q, u(q, f)).$$

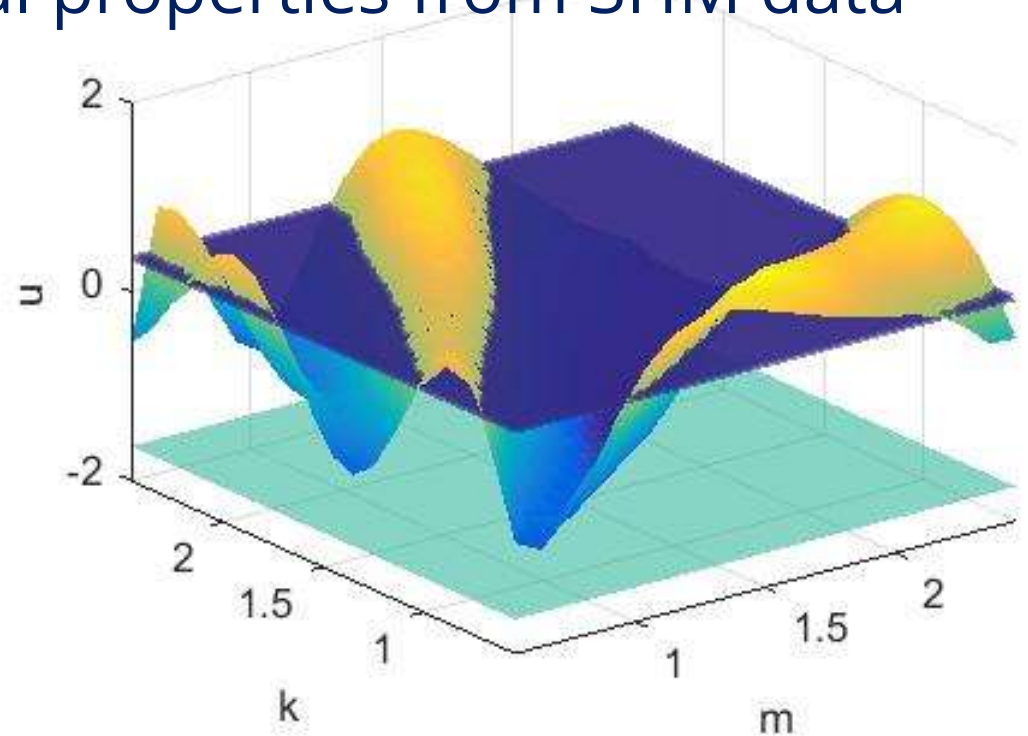
The actual measurement polluted by noise:

$$z(\omega) = y_t + \epsilon_m.$$

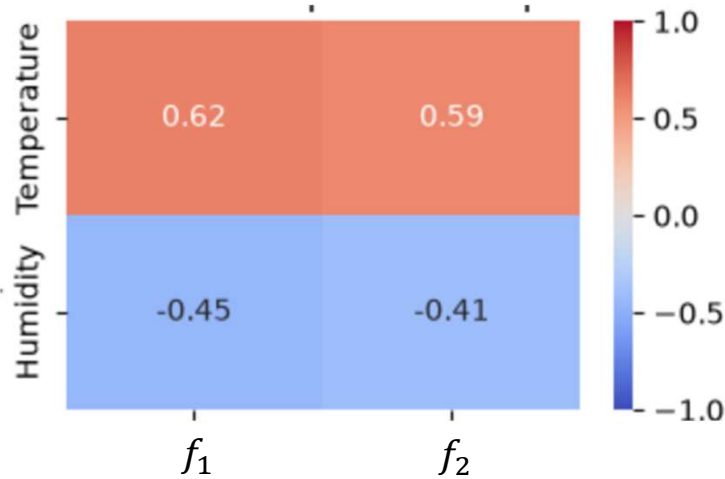
Bayesian framework

$$f_{q|Z} = \frac{f_{Z|Q} f_Q}{\int f_{Z|Q} f_Q dq}$$

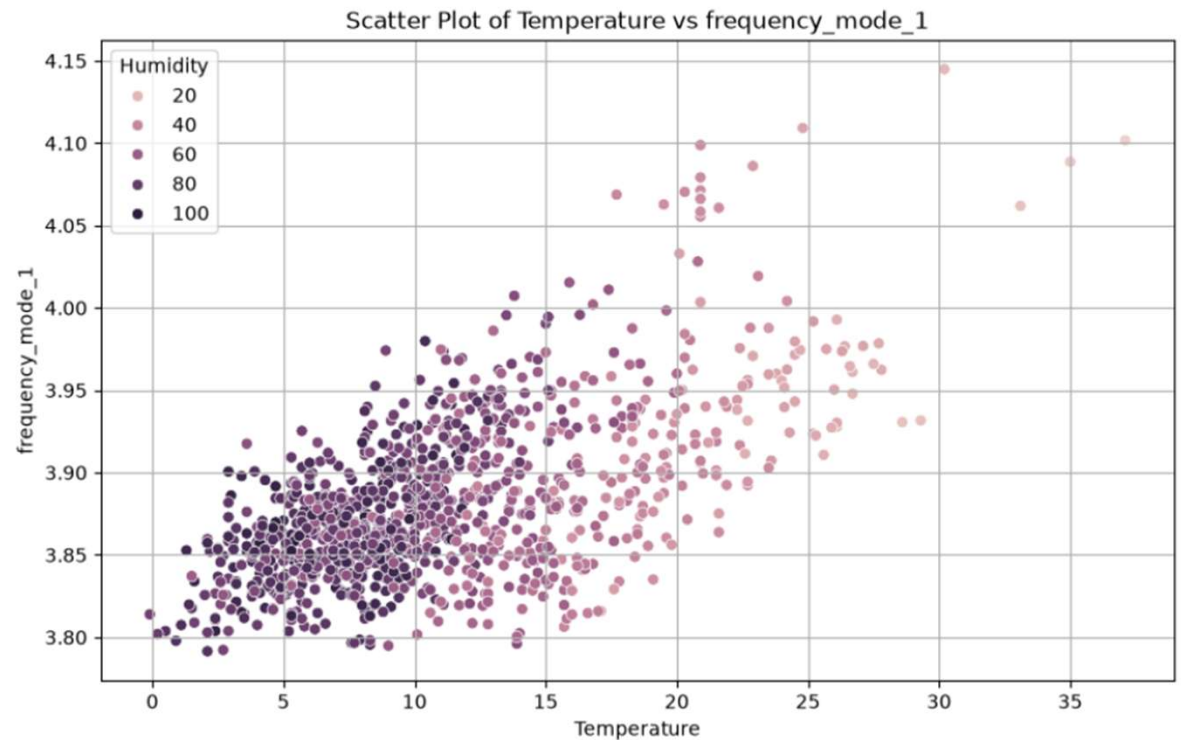
$$f_{Z|Q} = f_{\epsilon}(y - z)$$



Motivation: Noisy SHM data influenced by environmental condition



Frequency dependence on weather data of a cultural heritage building tower in Granada.



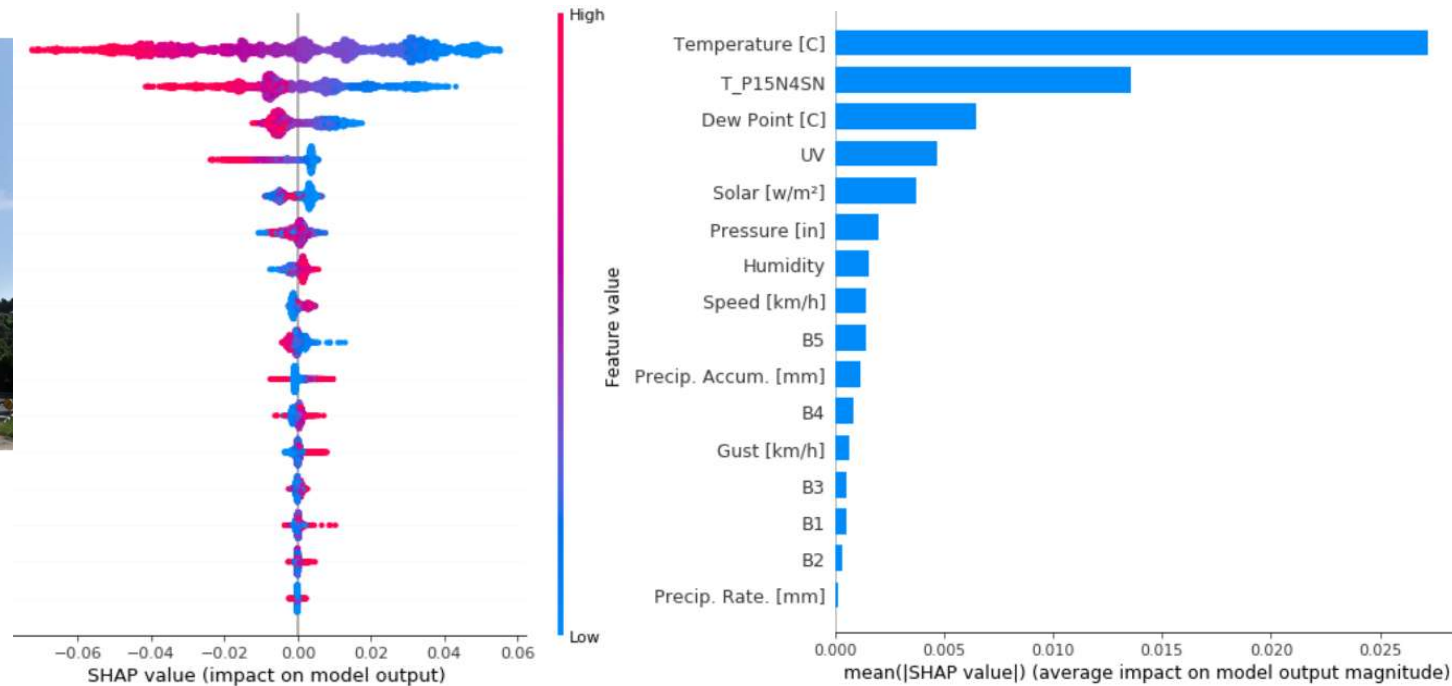
Approaches, Tools



Deterministic Approach to Subtract Environmental effects: Data-driven modelling, SHAPley Additive Feature attribution



- Ravbarkomanda high-way bridge



Environmental and traffic load effects on eigenfrequency of prestressed cable



Uncertainty-aware methods

Aleatoric



Randomness that is intrinsic to the phenomenon and cannot be reduced:

- Throwing a dice
- Measurement noise
- Unidentifiability

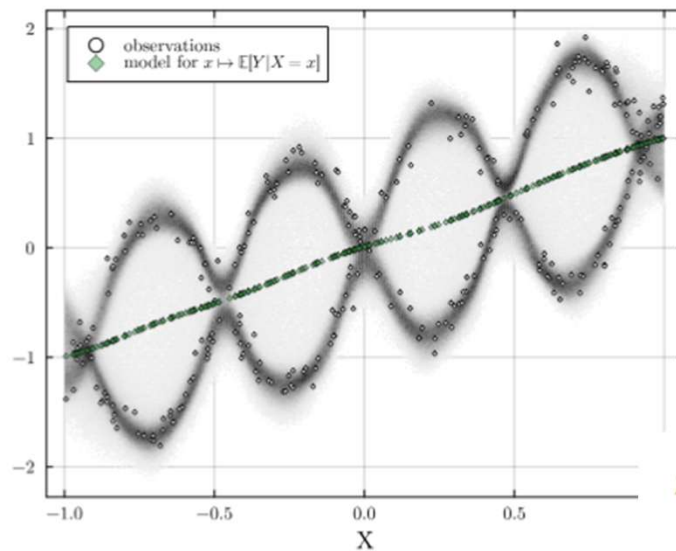
Instead of conditional expectation

$E[Y | X = x, D]$,
we want

$$p(y | X = x, D)$$

Distribution regression provide not only predictions but also a measure of confidence in those predictions

$$p(y|x) = \frac{1}{2} \mathcal{N}(y; \mu_1(x), \sigma^2) + \frac{1}{2} \mathcal{N}(y; \mu_2(x), \sigma^2)$$

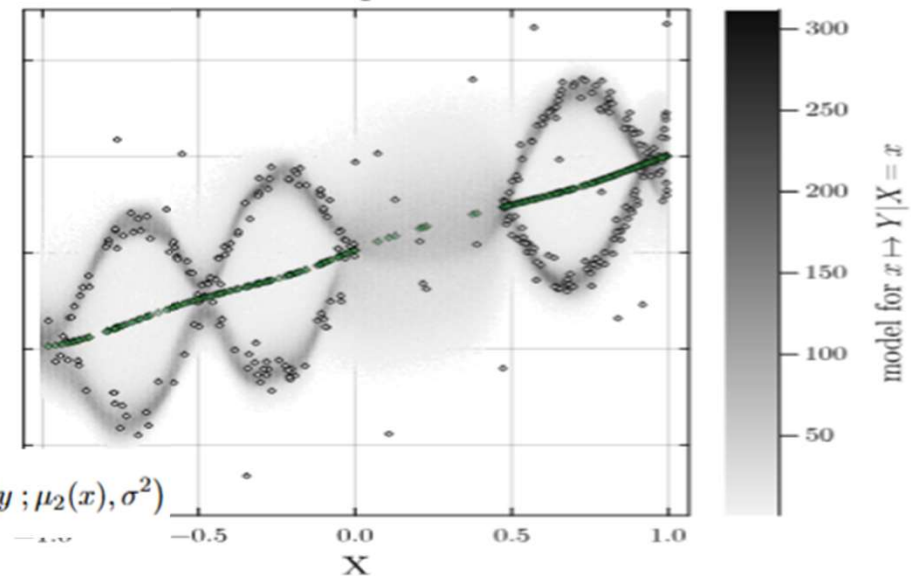


Epistemic

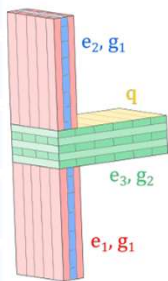
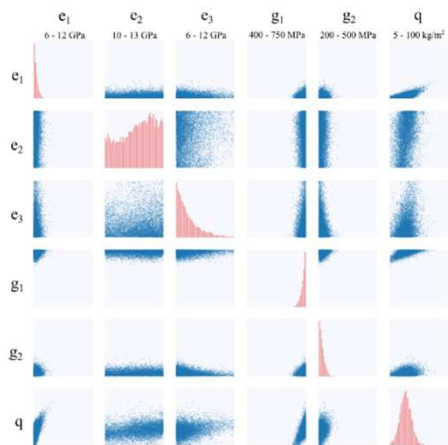
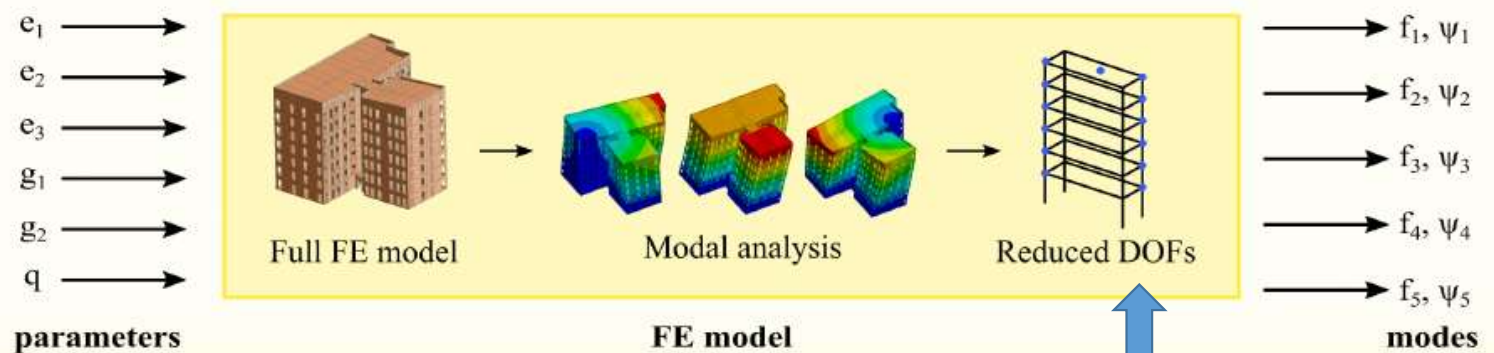


Reducible uncertainty, due to the lack of knowledge:

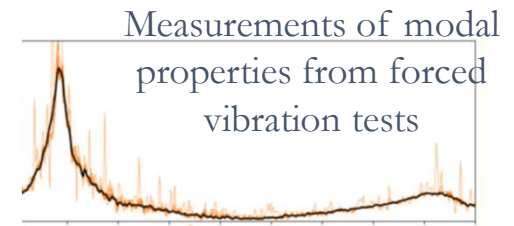
- Limited available observations for training model
- Not measurable input variables, uncertain material/geometric properties



Bayesian inversion with MCMC sampling of the posterior



Bayesian update of simulation model



$$f_{q|Z} = \frac{f_{Z|Q} f_Q}{\int f_{Z|Q} f_Q dq}$$

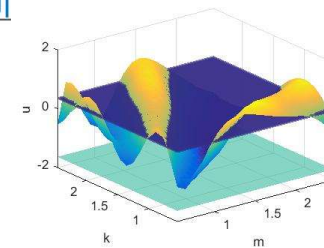
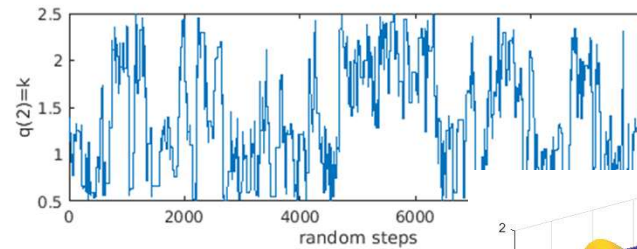
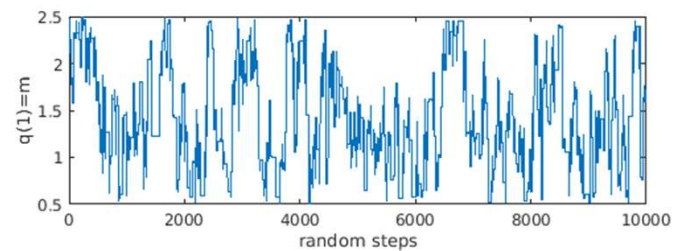
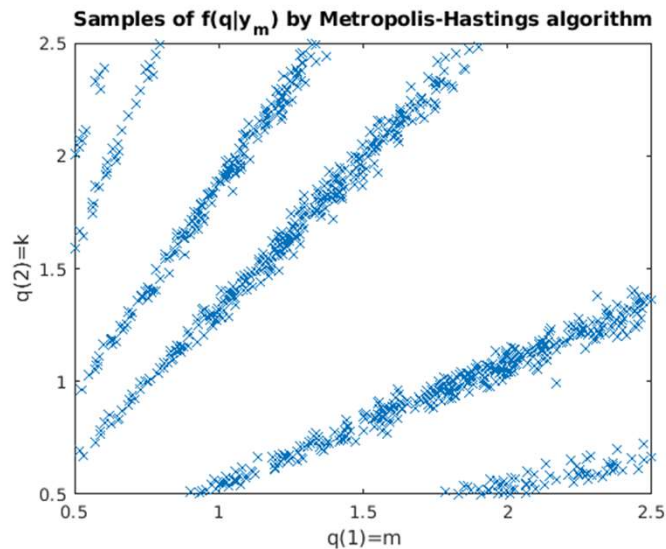
$$f_{Z|Q} = f_{\epsilon}(y - z)$$



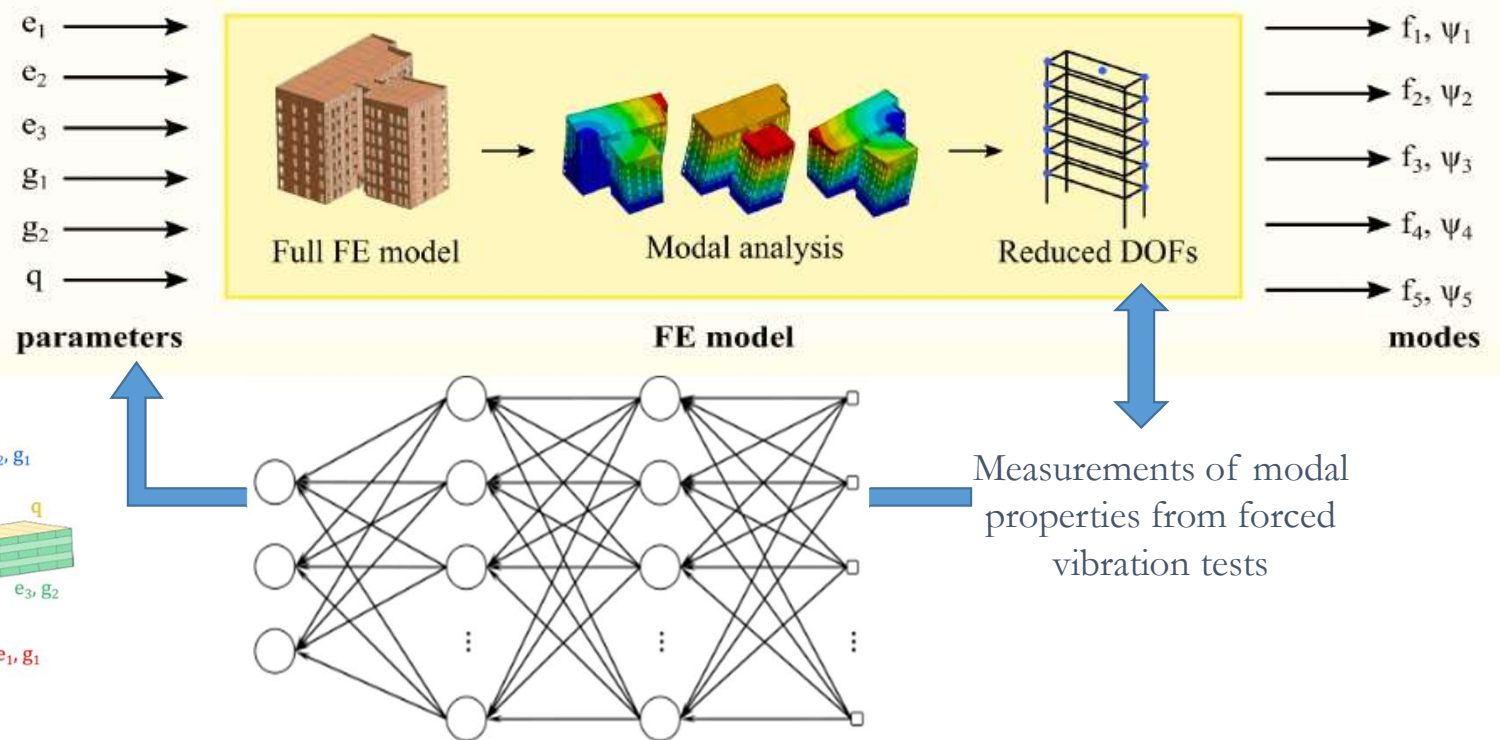
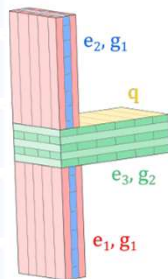
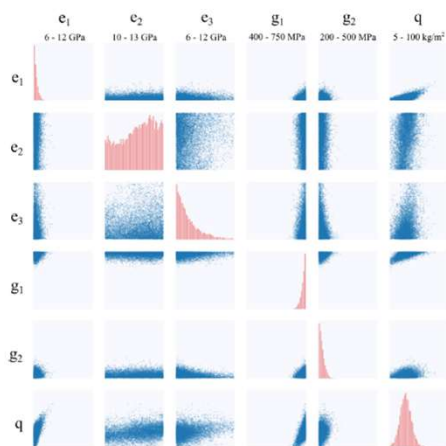
Bayesian inversion with MCMC sampling of the posterior

MCMC update of the spring example

https://ezander.github.io/ParameterAndFieldIdentification/param_identification/stochastic_estimators/Bayes_MCMC_spring.html



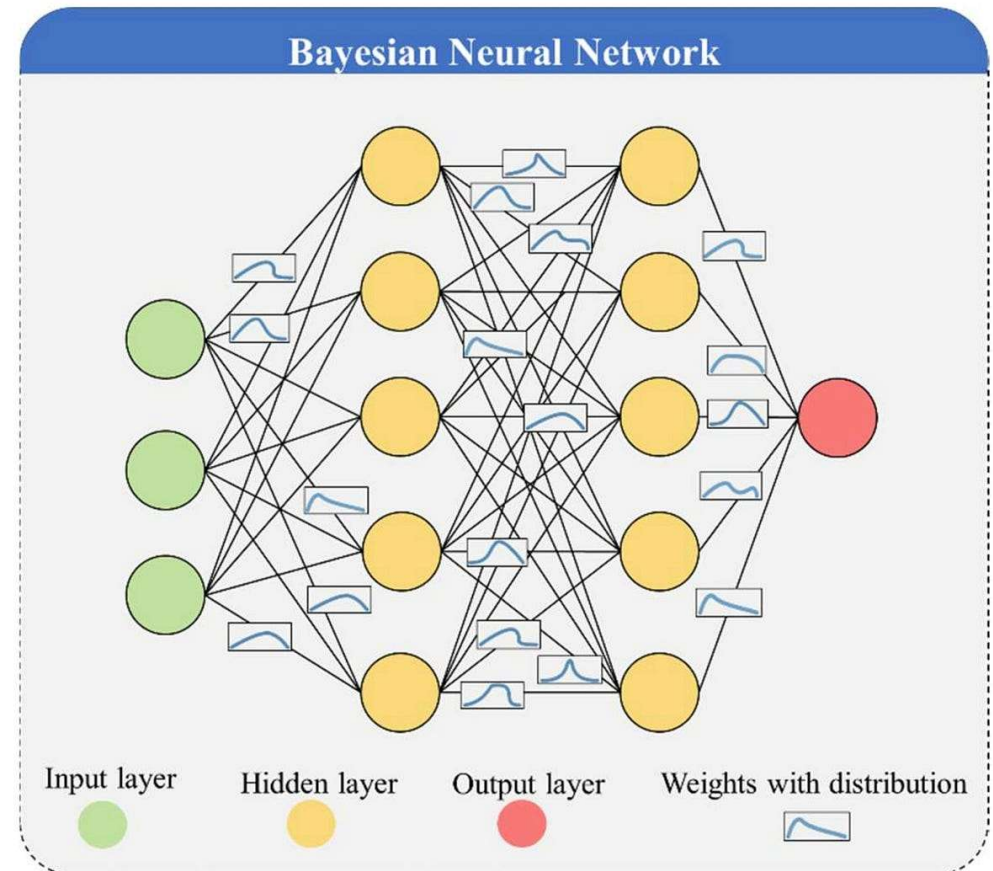
Probabilistic inverse modelling by distribution regression



Distribution Regression: Bayesian Neural Networks (BNNs)

- Weights (and biases) as probability distributions
- Bayesian inference of model parameters
- Computationally intensive
- Variational inference:
ELBO loss = **-Log-likelihood** of the data +
Kullback Liebler Divergence (**KLD**)
between the posterior and the prior
- Addresses epistemic uncertainty

[Magris, Iosifidis: Bayesian learning for neural networks: an algorithmic survey, Artificial Intelligence Review, 2023]



Source: https://miro.medium.com/v2/resize:fit:1200/1*ZmilGcmC6UmtN4uOQ1qsgg.jpeg



Distribution Regression: Mixture Density Networks (MDNs)

Conditional distribution is given by a Gaussian Mixture:

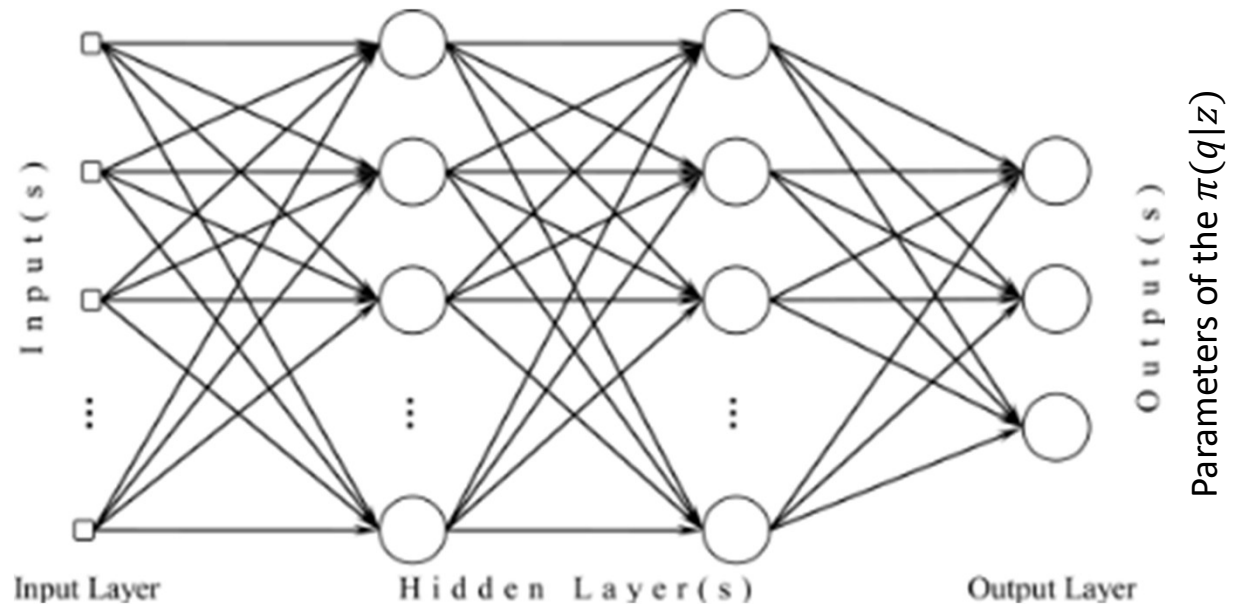
$$\pi(q|z) = \sum_{i=1}^m w_i(z) g_i(q|z) \quad \sum_{i=1}^m w_i = 1$$

Loss is the **negative loglikelihood**

$$-L_{(\omega,b)} = -\frac{1}{n} \sum_{j=1}^n \sum_{i=1}^m w_i(z_j) g_i(q_j|z_j)$$

$$z(\omega) = y_t + \epsilon_m$$

$$\Psi_{(\omega,b)}(z) = [\mu_1, \sigma_1, w_1, \mu_2, \sigma_2, w_2, \dots, \mu_m, \sigma_m, w_m]$$



w_i : mixing coefficients

g_i : conditional density (Gaussian kernel functions)



Distribution Regression: CRPS-based distribution regression

Noise $\varepsilon \sim N(0, I)$ is injected into the network; the network maps $(y, \varepsilon) \mapsto p$. Multiple draws of ε give samples approximating the posterior.

Novel unbiased empirical approximation of the CRPS:

$$CRPS'(F, q) = \int_R (F(r) - \mathbf{1}(q \leq r))^2 dr$$

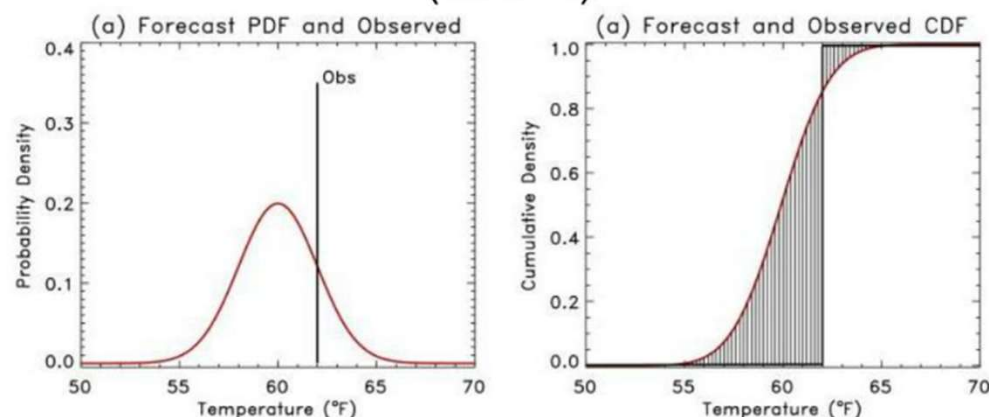
$$CRPS(F, q) = E |Q - q| - \frac{1}{2} E |Q - Q'|, \quad Q, Q' \sim F$$

F : CDF of predicted distribution
 q : target label

The model is trained by minimizing the unbiased version of the empirical Continuous Ranked Probability Score (CRPS):

$$CRPS'(q) = \frac{1}{n} \sum_{i=1}^n |q - \hat{q}_i| - \frac{1}{2n(n-1)} \sum_{i=1}^n \sum_{j=1}^n |\hat{q}_i - \hat{q}_j|$$

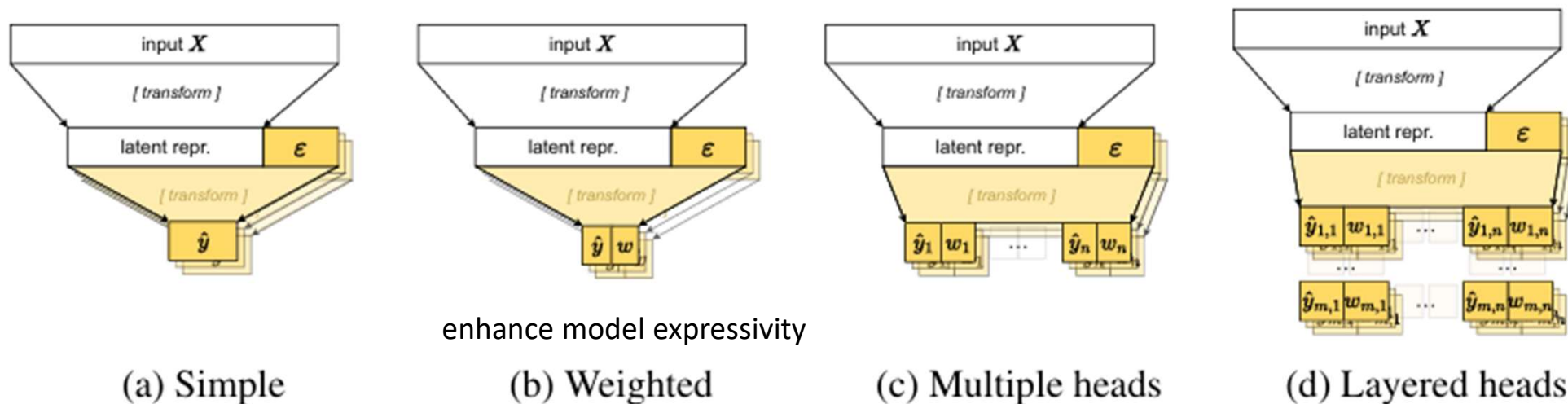
Strictly proper scoring rule: global optimum coincides with the true target distribution — no parametric assumption required, capable of learning arbitrary probability distributions.



Distribution Regression: CRPS-based distribution regression

Kelen DM, Jung Á, Benczur AA. Quantification vs. Reduction: On Evaluating Regression Uncertainty. Learning meets Sampling. ICLR 2025 workshop

Kelen DM, Jung Á, Kersch P, Benczur AA. Distribution-free Data Uncertainty for Neural Network Regression. ICLR 2025.



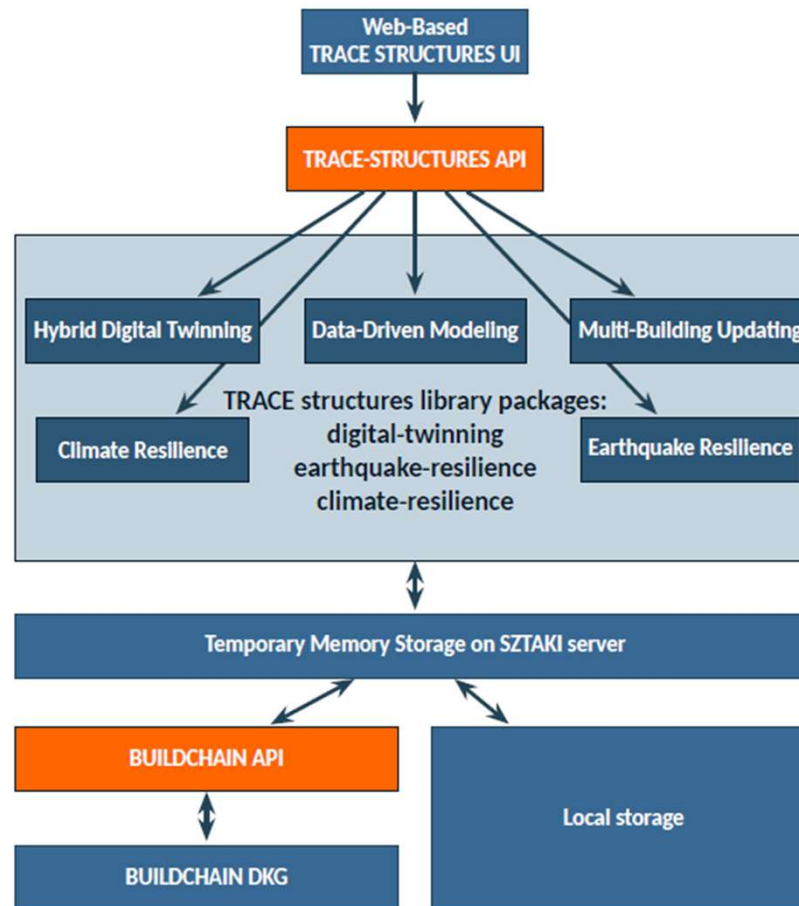
[GitHub - proto-n/torch-naut: PyTorch Nonparametric Aleatoric Uncertainty Modeling Toolkit · GitHub](https://github.com/proto-n/torch-naut)
<https://github.com/proto-n/torch-naut/tree/main>



TRACE STRUCTURE UI: some tools that are available without coding expertise

API for
Twinning,
Resilience, and
AI-driven
Computational
Engineering for
Building
Structures

<https://buildchain.ilab.sztaki.hu/>

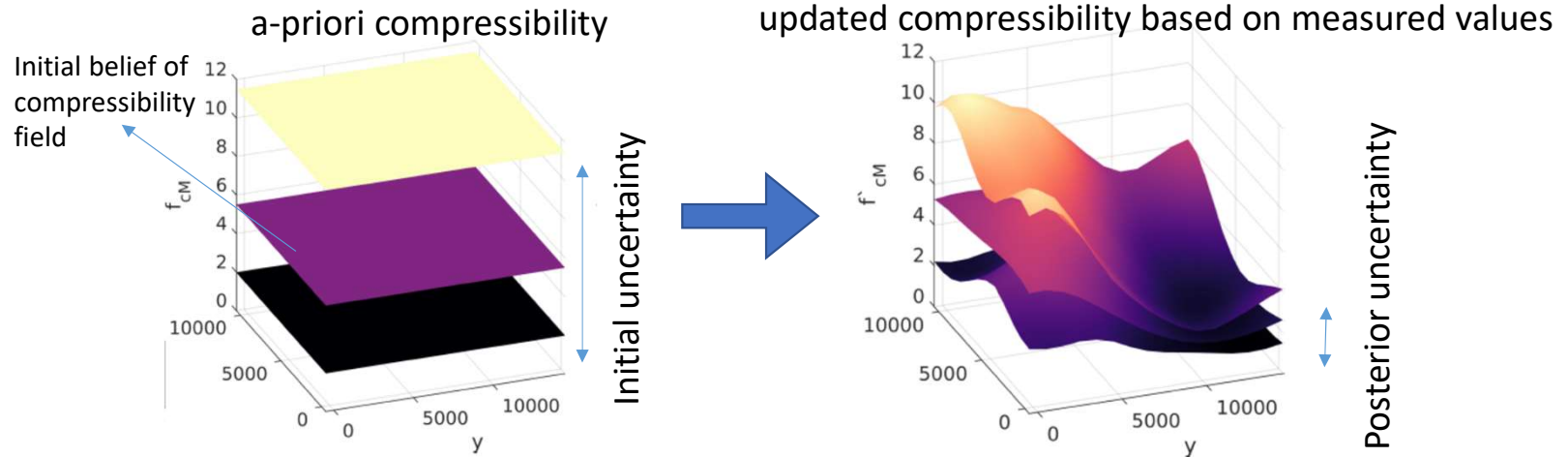
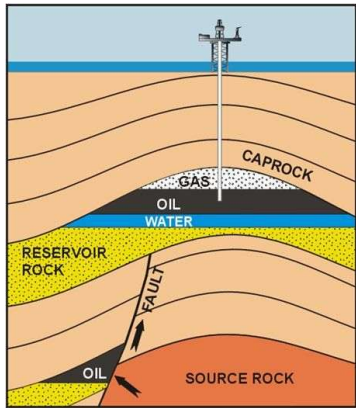


Challenges, Limitations



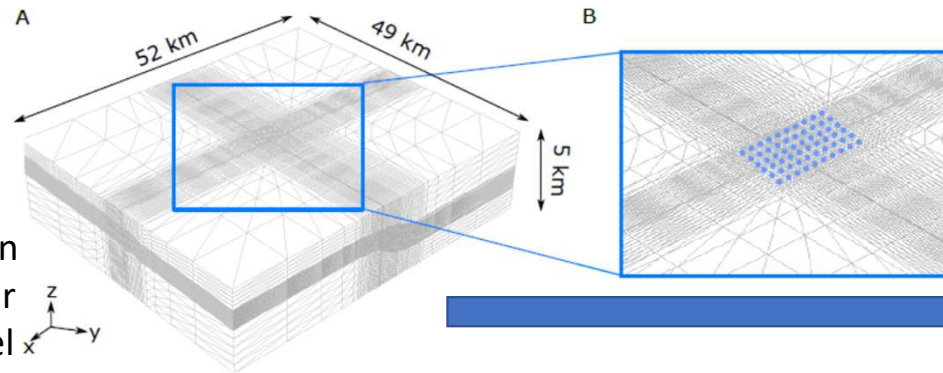
Challenges1: Curse of dimensionality, field updating

Compressibility field identification to preserve safety of a gas platform



Uncertain scaling field of the compressibility $f_{CM}(x, y)$ under the gas platform

FEM simulation with non-linear material model



measured seabed displacement

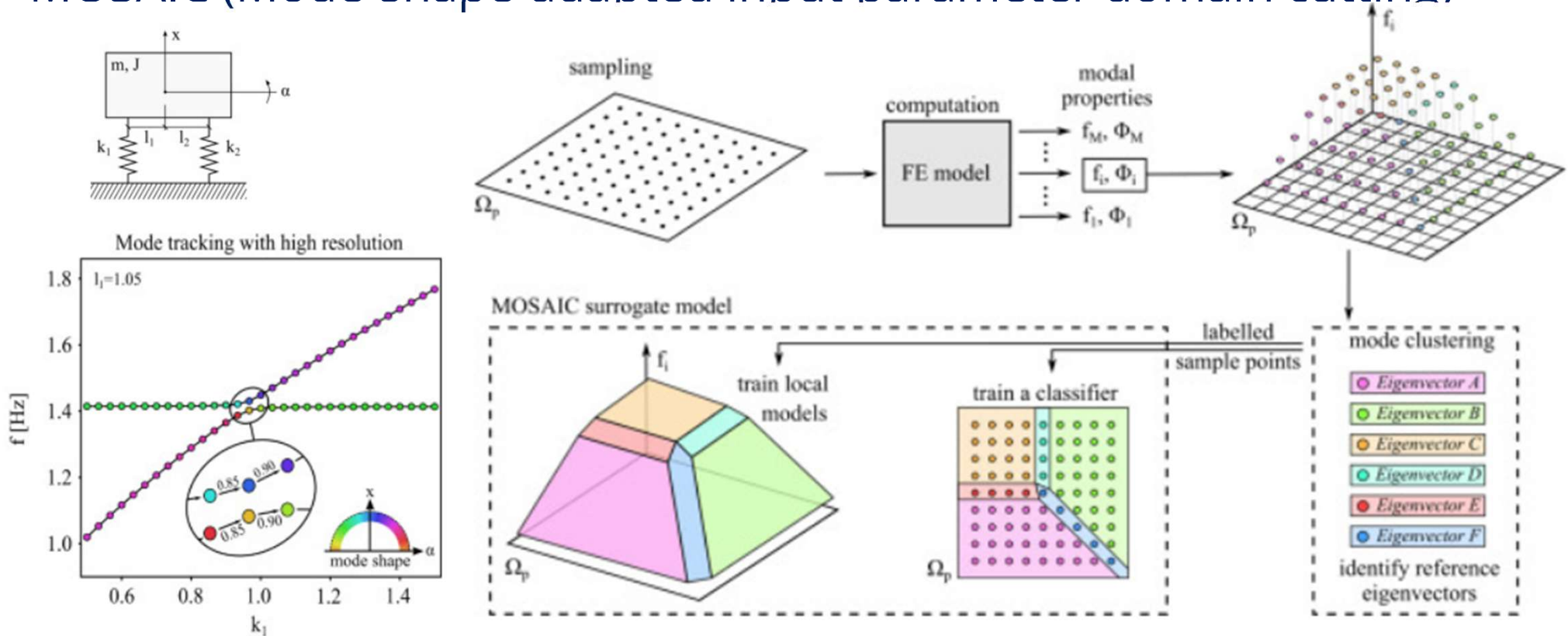
Stochastic FEM computation of seabed displacement

Posterior uncertainty

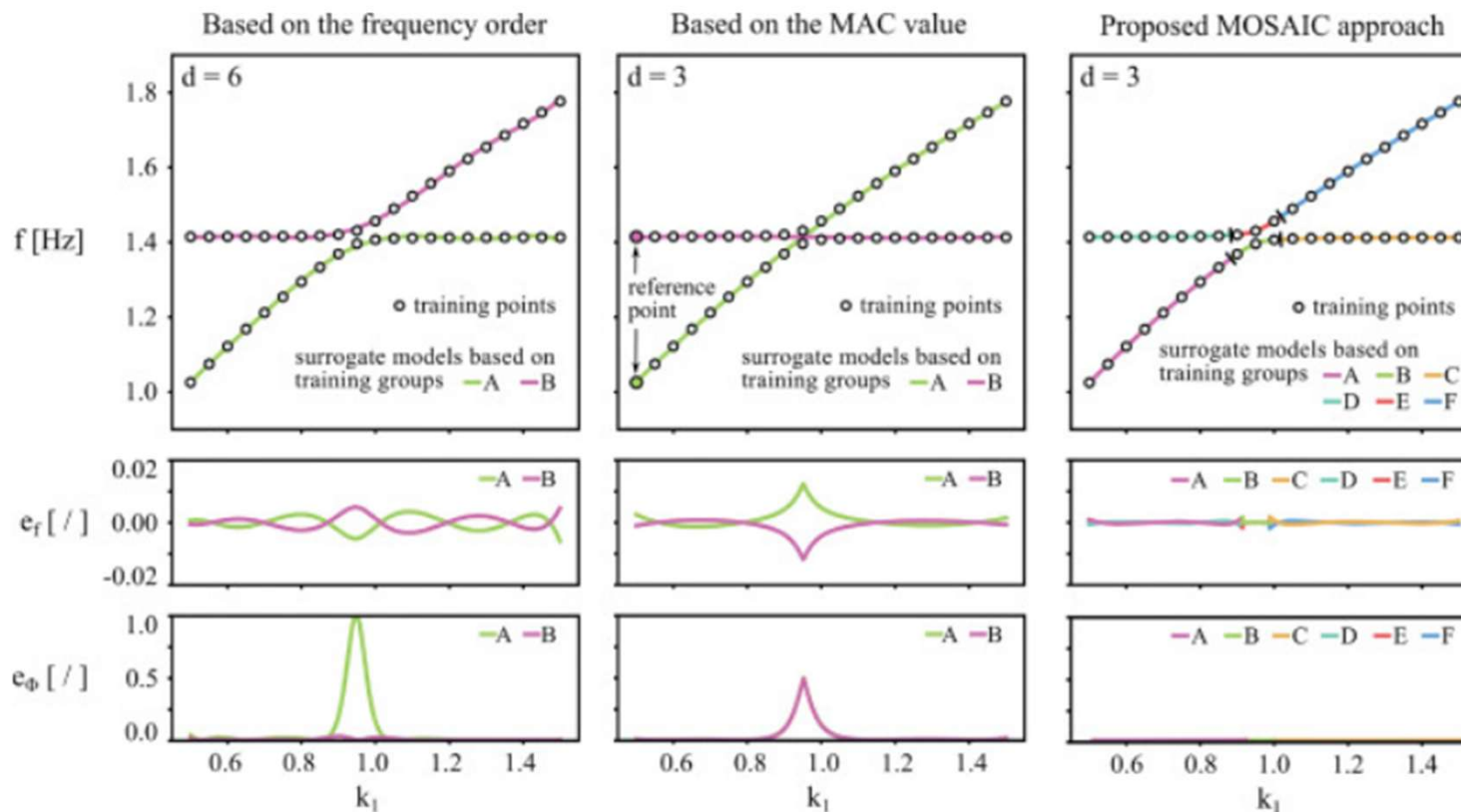


Challenge 2: Non-smooth behavior when measurements are modal properties

MOSAIC (Mode-shape-adapted input parameter domain cutting)



MOSAIC (Mode-shape-adapted input parameter domain cutting)





Challenge 3: Knowledge and data sharing decentralized knowledge graphs – the BUILDCHAIN project

A smarter, transparent and more resilient and sustainable built environment:

Build a Knowledge Base, via a **Decentralized Knowledge Graph**, that can be used to trace all activities related to the overall life-cycle of buildings.

- **Integration** of trusted, traceable, and transparent data
- **Marketplace**, where various actors can share their offers, including their quality certificates and credentials, and where it would be possible to log and trace every information, activity, and change, and use the knowledge to improve sustainability.

The creation of a **trusted data ecosystem economy** around such high-quality data will significantly **improve the offered services and products in the Building Life-Cycle sector**.

Interconnectivity

Data and AI fusion

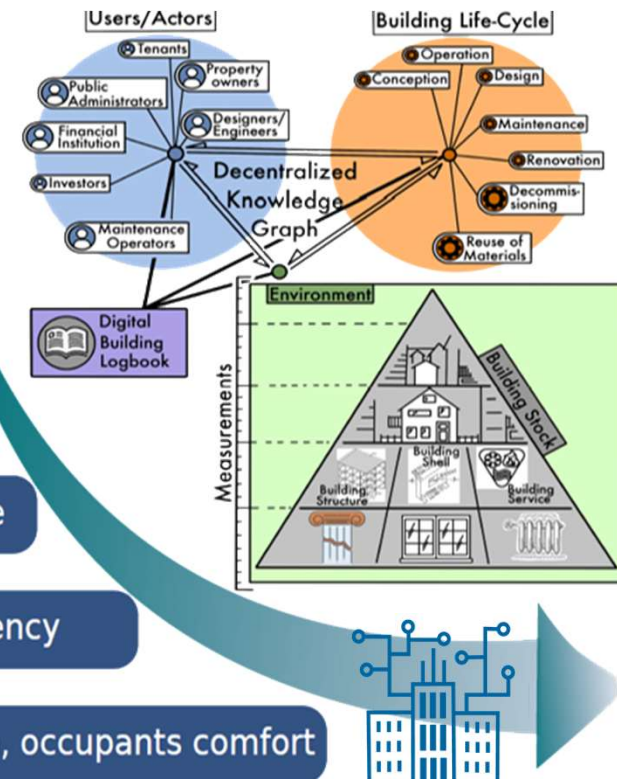
New collaboration & services

New technologies & innovation

Data & knowledge market place

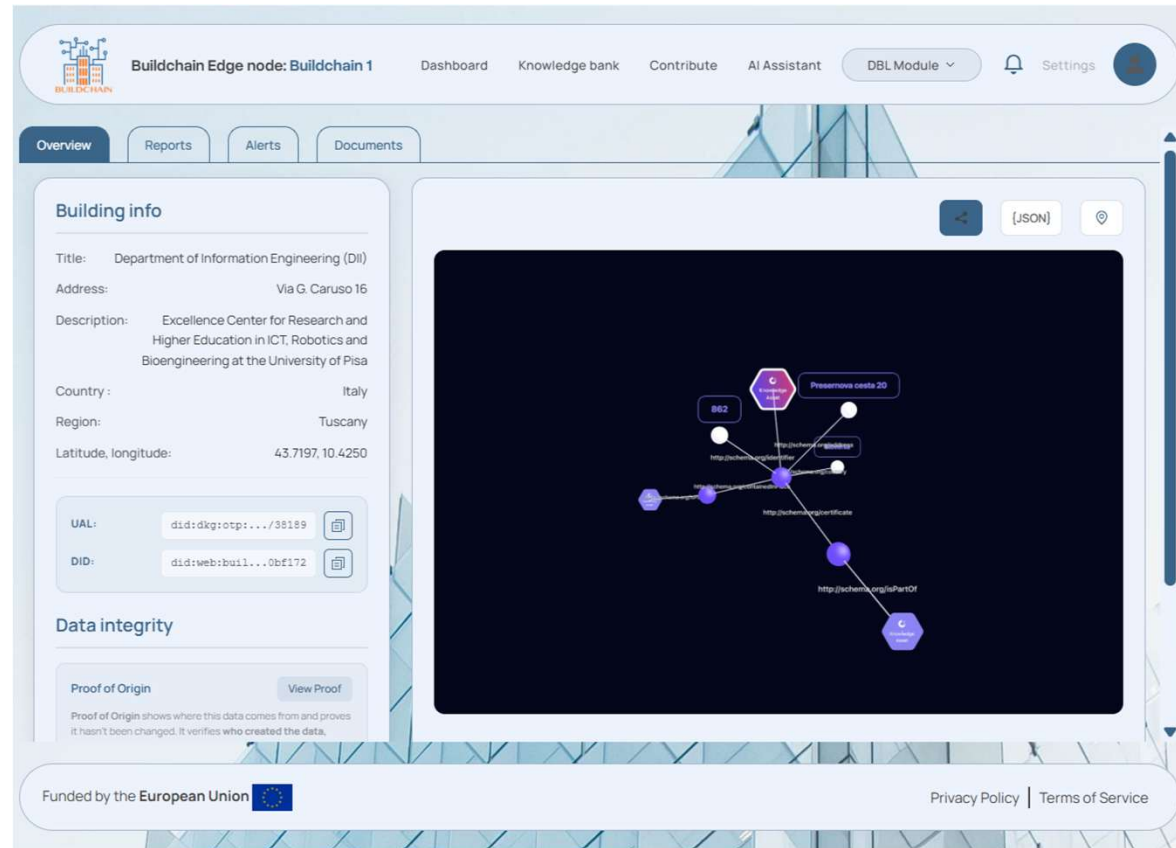
Sustainability & energy efficiency

Improved safety, health, occupants comfort



Challenge 3: Knowledge and data sharing

Structured knowledge assets organized in a graph



Challenge 3: Knowledge and data sharing

AI-tools for easy data query

Search

Chat



Can you tell me more about the monitoring system?



The building is equipped with a comprehensive monitoring system that includes 6 seismic acceleration sensors. These sensors are METRA KB12VD models, which have a sensitivity of 10000mV/g and can measure a range between $\pm 0.5g$ to $\pm 5g$. They are strategically installed in the central tower of the building.

These sensors continuously collect acceleration data in the two main directions (x and y) of the building. The collected data is stored in a database and further processed using Operational Modal Analysis. This analysis helps identify the natural vibration modes of the building, which are closely related to the building's state of health. Any changes in these vibration modes can indicate the presence of damage in the building.



Tell me information about this building?

SOURCE KNOWLEDGE ASSETS



Hospital Real Building
Issuer: [Trace Labs](#)

View source content



Frequency reader daytime 19-...
Issuer: [Trace Labs](#)

View source content



Frequency reader nighttime 19-...
Issuer: [Trace Labs](#)

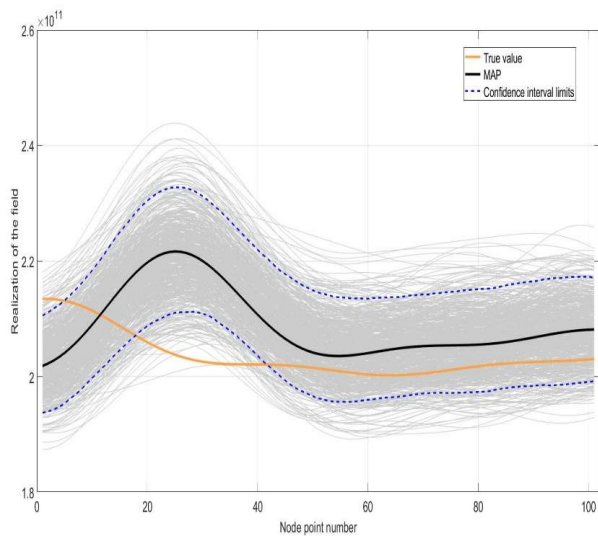
View source content



Limitation: identifiability

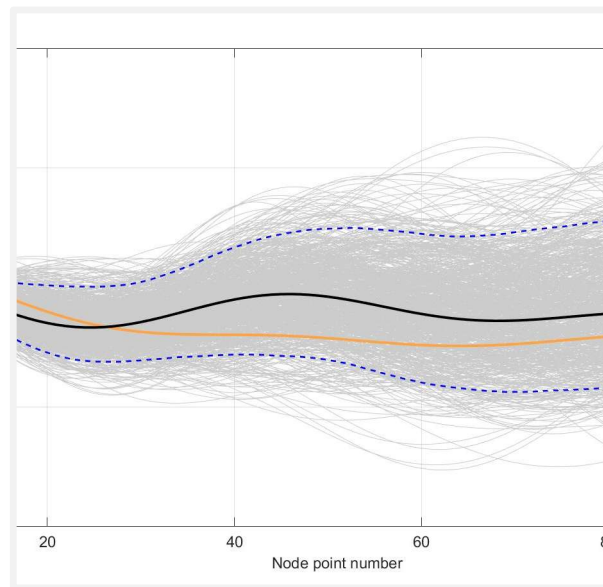
Dynamic experiment

– measuring **mode shapes**



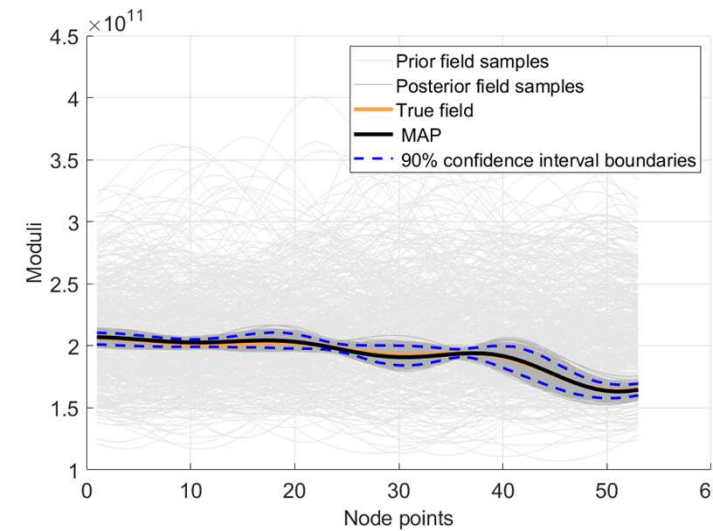
Static experiment

– measuring **displacements**

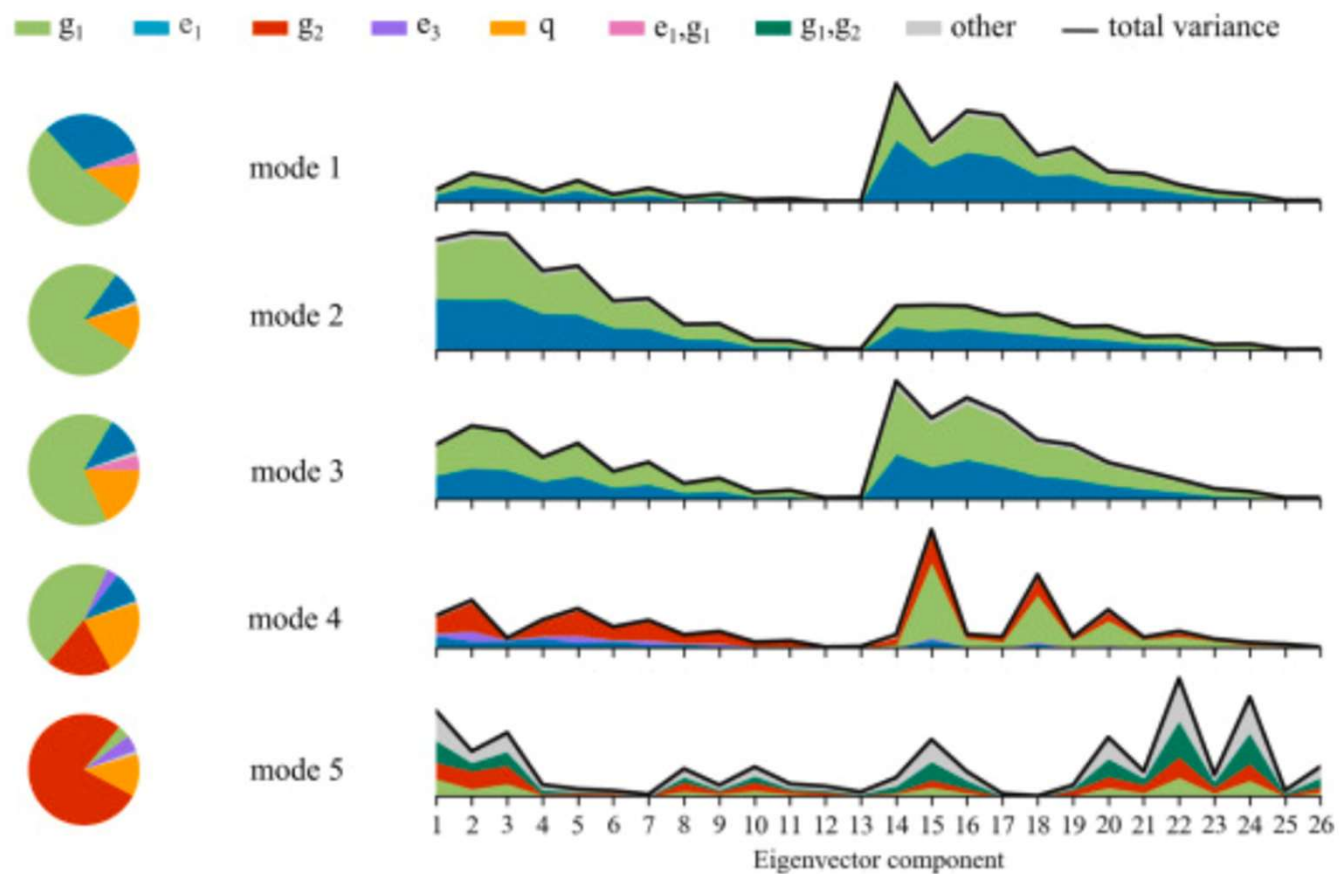


Static experiment

– measuring **strains**



Limitation: identifiability



Summary



Summary

Uncertainty-aware SHM: opportunities and challenges

Uncertainty-aware methods are recommended for both **parameter identification** and **modelling environmental effects** in SHM data.

Several uncertainty-aware modelling tools are already available:

- Bayesian inversion with **surrogate models + MCMC**

- Distribution regression: **Bayesian NNs, MDNs, CRPS-based methods**

Key challenges for practical deployment:

- Systematic error modelling** and robust **generalization** for design-method updating

- Modal measurements:** widespread in SHM, but affected by **mode matching** and **aliasing errors**

- Curse of dimensionality**

- Data availability, structure, and reusability**

- Non-identifiability**, limiting exact parameter identification



Some references

- **BAYESIAN INVERSION:** Friedman N, Zoccarato C. et al. "A Worked-out Example of Surrogate-based Bayesian Parameter and Field Identification Methods." Bayesian Methods for the Analysis of Engineering Systems; Chiachio Ruano, J., Chiachio Ruano, M., Sankararaman, S., Eds (2021).
- **BAYESIAN INVERSION:** Kurent N., Friedman N. et al: Bayesian updating of tall timber building model using modal data, Engineering Structures, in review
- **BAYESIAN INVERSION:** Marsili F., Croce P. et al. "Seismic reliability assessment of a concrete water tank based on the Bayesian updating of the finite element model." ASCE-ASME J Risk and Uncert in Engrg Sys Part B Mech Engrg 3.2 (2017).
- **BAYESIAN INVERSION:** Landi F., Marsili F. et al. gPCE-Based Stochastic Inverse Methods: A Benchmark Study from a Civil Engineer's Perspective. Infrastructures, 6(11), 158. (2021)
- **Mean Squared Estimator:** Zander E., Friedman N., & Matthies H. G. Parameter Identification Based on Conditional Expectation., Bayesian Methods for the Analysis of Engineering Systems; Chiachio Ruano, J., Chiachio Ruano, M., Sankararaman, S., Eds (2021).
- **MOSAIC:** Kurent, Blaž ; Popovics, Bence ; Brank, Boštjan ; Friedman, Noémi: A novel approach to surrogate modelling of modal properties: Mode-shape-adapted input parameter domain cutting MECHANICAL SYSTEMS AND SIGNAL PROCESSING 240 p. 113381 Paper: 113381 (2025)
- **BUILDCHAIN:** Bevilacqua, M G and Caroti, G and Croce, P and Friedman, Noémi and Landi, F and Piemonte, A (2023) *The Horizon Europe "BUILDCHAIN" Project Presentation: Trustworthy Building Life-Cycle Knowledge Graph for Sustainability and Energy Efficiency*. SCIRES-IT: SCIENTIFIC RESEARCH AND INFORMATION TECHNOLOGY, 13 (2). pp. 137-150. ISSN 2239-4303 [10.2423/i22394303v13n2p137](#)
- **BUILDCHAIN:** Petar Kochovski, Vlado Stankovski et al: The BUILDCHAIN platform for trustable knowledge management in construction– Use Cases, Architecture and Design, Central Europe towards Sustainable Building 2025, [CESB25 Proceedings Technical.pdf](#)
- **BUILDCHAIN:** Friedman, Noémi ✉ ; Kurent, Blaž et al. [Digital Twinning for Enhancing Transparency in Sustainability, Safety, Health and Serviceability of Tall Timber Structures](#) In: WCTE - WCTE (szerk.) [World Conference on Timber Engineering 2025: Advancing Timber for the Future Built Environment](#) Red Hook (NY), USA: Curran Associates (2025) 5,799 p. pp. 2260-2269. , 10 p.
- **CAUSAL EXPLANATIONS:** DM Kelen, P Kersch, AA Benczúr. [Causal Explanations for Performance in Radio Networks](#). AI6G Workshop 2022
- **CAUSAL EXPLANATIONS:** DM Kelen, M Petreczky, P Kersch, AA Benczúr Theoretical evaluation of asymmetric shapley values for root-cause analysis. ICDM 2023.
- **DISTRIBUTION REGRESSION:** DM Kelen, Jung Á, Benczur AA. Quantification vs. Reduction: On Evaluating Regression Uncertainty. Learning meets Sampling. ICLR 2025 workshop
- **DISTRIBUTION REGRESSION:** Kelen DM, Jung Á, Kersch P, Benczur AA. Distribution-free Data Uncertainty for Neural Network Regression. ICLR 2025.

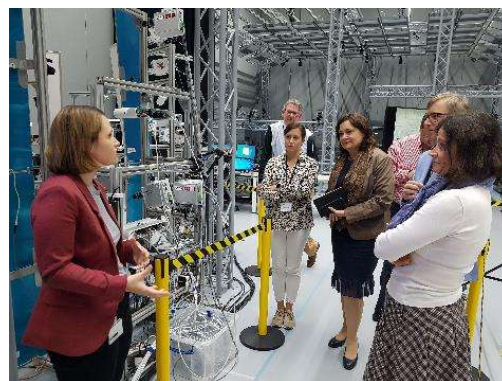
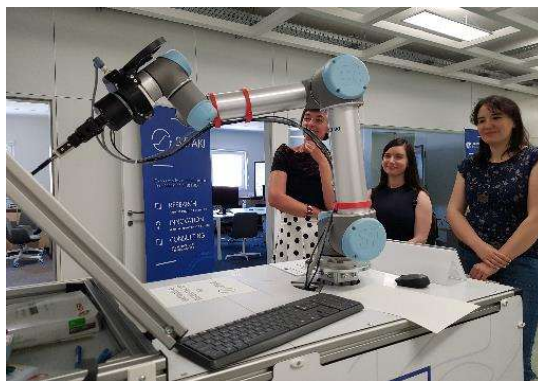


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Thank you!



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